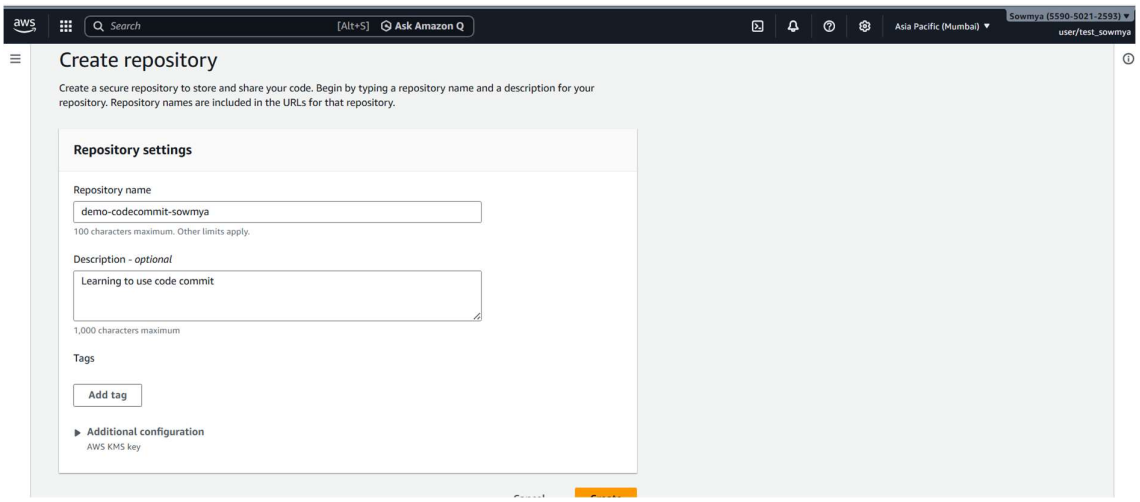
CODE COMMIT

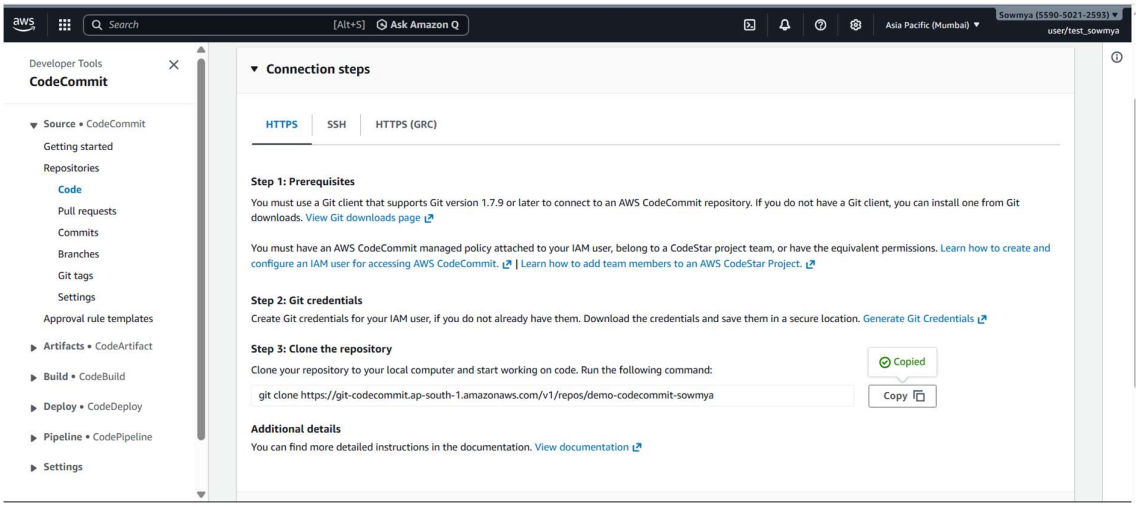
Step 1 : Login to your AWS IAM user before that give AWSCodecommitfullaccess to the IAM user through your root account.

Step 2: Search for Code Commit and click on create repository.



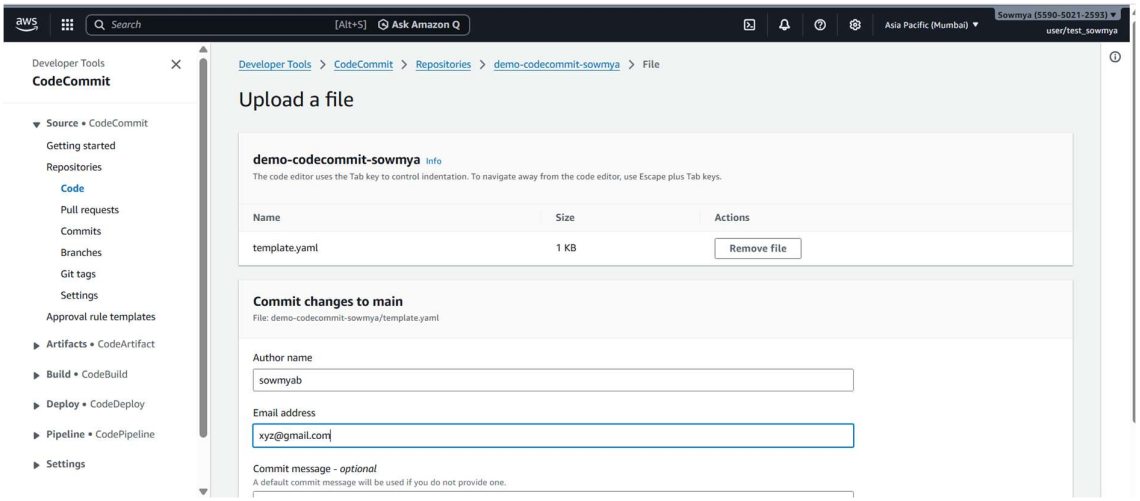
Step 3 : You will see below screen after u click on create. Copy the clone repository URL.

So that we will try to clone the repository through ubuntu terminal.



Step 4: Before cloning, scroll down and you will see add file button->select upload file.

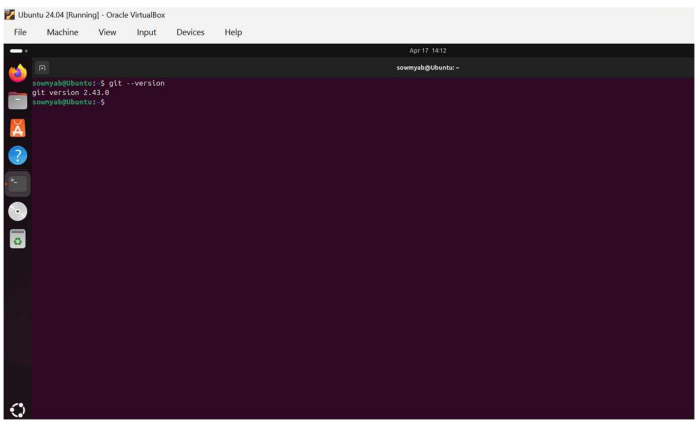
Here you can give your email address and then click on commit changes.



Step 5: Open the ubuntu terminal and install git using below command

sudo apt install git -y

To verify use git –version



Step 6: Now use git clone command to clone the repository. Before that generate SSH key in ubutu with following command.

ssh-keygen -t rsa -b 2048

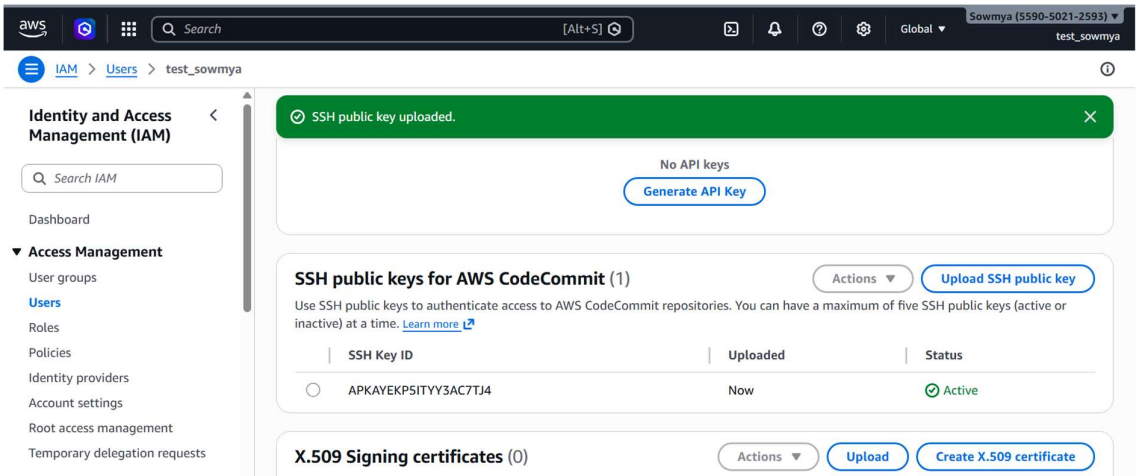
```
sowmyab@Ubuntu:~$ git --version
git version 2.43.0
sowmyab@Ubuntu:~$ ssh-keygen -t rsa -b 2048
Generating public/private rsa key pair.
Enter file in which to save the key (/home/sowmyab/.ssh/id_rsa): /home/sowmyab/.ssh/id_rsa
Enter passphrase (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /home/sowmyab/.ssh/id_rsa
Your public key has been saved in /home/sowmyab/.ssh/id_rsa.pub
The key fingerprint is:
SHA256:KLwa7n6FqDJkz02CJkVQbEFDA3aDZIflXEL3rUPcGdI sowmyab@Ubuntu
The key's randomart image is:
+---[RSA 2048]----+
|+@%O o ...       |
|ooOo= o +Eo      |
| o o   + +       |
|  ..  ...        |
| ...o..oS        |
|.=...+. .        |
|=.+ =.           |
|+. =..           |
|.+=.             |
+----[SHA256]-----+
sowmyab@Ubuntu:~$ cat ~/.ssh/id_rsa.pub
```

Step 7: Now use cat command copy the secret key.

cat ~/.ssh/id_rsa.pub

Step 8 : Open IAM and select the user and go to security credentials->upload the ssh key generated.

After uploading you will see as below image.



Step 9 :Use below commands in your ubuntu terminal

nano ~/.ssh/config

Host git-codecommit.*.amazonaws.com User APKAEIBAERJR2EXAMPLE IdentityFile ~/.ssh/id_rsa

APKAEIBAERJR2EXAMPLE – here, paste the ssh key id from your console

The SSH config file tells the system which private key and SSH Key ID to use when connecting to AWS CodeCommit during operations like git clone.

Save file:

- Press Ctrl + X
- Press Y
- Press Enter

After that give permissions to file with below commands

chmod 600 ~/.ssh/config

chmod 600 ~/.ssh/id_rsa

ssh git-codecommit.ap-south-1.amazonaws.com

It will if u want to connect ->give yes

We use chmod 600 to restrict access to the SSH private key and config file, because SSH requires strict permissions for security and will reject keys that are accessible by others

Step 10: Now clone the repository

git clone ssh://git-codecommit.ap-south-1.amazonaws.com/v1/repos/demo-codecommit-sowmya

Go to AWS console->Code commit->select the repository->copy ssh url

Step 11 : To verify if cloning was successful or not use below commands

cd demo-codecommit-sowmya

ls

```
sowmyab@Ubuntu:~$ cd demo-codecommit-sowmya
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ ls
template.yaml
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
sowmyab@Ubuntu:~/demo-codecommit-sowmya$
```

Step 12 : Let's create a file in the repository. Before that use below command

git config --global user.name "Your Name"

git config --global user.email "your-email@example.com"

Git requires user.name and user.email to identify the author of each commit. This metadata is stored in commit history and helps track changes across the repository.

echo "My first CodeCommit project" > README.md creates a file and writes text into it

git add README.md - Moves the file to staging area

git commit -m "Added README file" - Saves changes permanently in local repo

git push - Uploads commits to remote repository

git push origin main - Pushes main branch to remote named origin

Git has 3 layers:

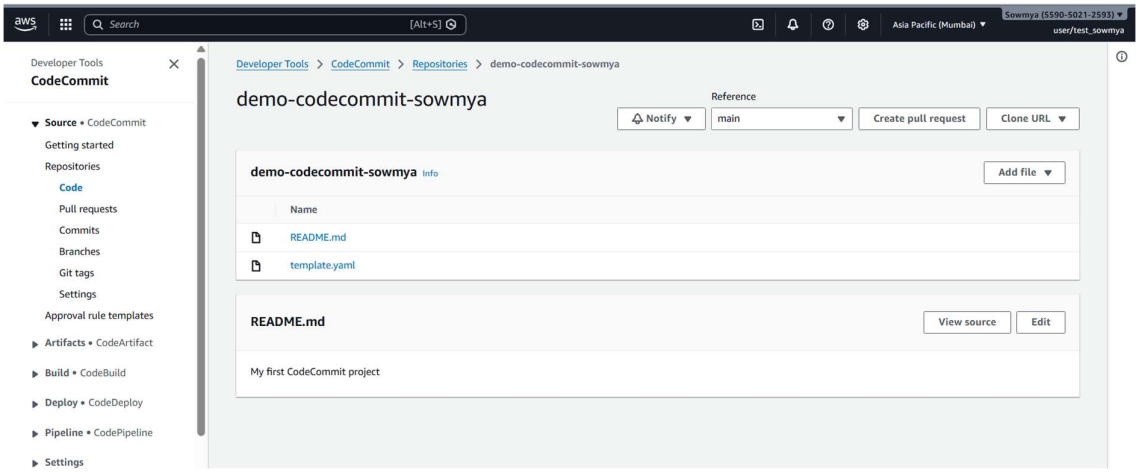
1. Working directory
2. Staging area
3. Repository

This command moves file from:
Working directory → Staging area

Why we use it

- Git doesn't track files automatically
- You explicitly tell Git:

Include this file in next commit

```
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ echo "My first CodeCommit project" > README.md
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ git add README.md
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ git commit -m "Added README file"
[main 6874736] Added README file
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 318 bytes | 318.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
remote: Validating objects: 100%
To ssh://git-codecommit.ap-south-1.amazonaws.com/v1/repos/demo-codecommit-sowmya
   eef71bd..6874736  main -> main
sowmyab@Ubuntu:~/demo-codecommit-sowmya$ git push origin main
Everything up-to-date
```

Step13 : Now open AWS console go to repository you will now see a README.md file too.